

Dominated convergence fails for infinite-rank linear conditional expectation

Candidate full counterexample to the open problem in arXiv:2008.12070

12 August 2026

Status: candidate full counterexample, likely valid. There exist a probability space, a centered random variable $V \in L^2(\Omega; \ell^2)$ whose covariance has infinite rank, scalar $U_k \in L^2(\Omega)$, and $Y \in L^2(\Omega)$ such that

$$U_k \longrightarrow 0 \quad \text{almost surely,} \quad |U_k| \leq Y \quad (k \geq 1),$$

but

$$\mathbb{E}^{\mathcal{A}}[U_k \mid V] \not\longrightarrow 0 \quad \text{almost surely.}$$

Thus the infinite-rank extension asked for in Remark 4.6 of Klebanov, Sprungk, and Sullivan is false. The construction realizes a graph-subspace orthogonal projection as an LCE and feeds it the tails of a classical Menshov orthogonal series that converges in L^2 but diverges pointwise.

1 The exact open problem

For scalar $U \in L^2(\Omega)$ and Hilbert-valued $V \in L^2(\Omega; \mathcal{H})$, the linear conditional expectation (LCE) is the L^2 -orthogonal projection

$$\mathbb{E}^{\mathcal{A}}[U \mid V] = P_{\{b + \langle h, V \rangle : b \in \mathbb{R}, h \in \mathcal{H}\}} U. \quad (1)$$

The source proves an L^2 -dominated convergence theorem when the covariance C_V has finite rank. It asks whether the result remains true at infinite rank under the stronger hypothesis

$$U_k \rightarrow U \text{ a.s.,} \quad |U_k| \leq Y \text{ for one } Y \in L^2. \quad (2)$$

The complete source page containing the question is reproduced below [\[1\]](#).

the conventional assumption $Y \in L^1(\mathbb{P})$ (a counterexample to the sufficiency of the latter assumption is provided in [Theorem 4.7\(g\)](#)). On the other hand, (β) is a particularly weak assumption, too weak for analogous statements on the (regular) conditional expectation $\mathbb{E}[\cdot|V]$ in place of $\mathbb{E}^\mathbb{A}[\cdot|V]$.

So far, we could only prove the DCT under the condition that C_V has finite rank. Note that counterexamples can easily be constructed (see below) if one only assumes (β) . However, the validity of the DCT under (α) for C_V of infinite rank remains an open problem. One obstacle here is the missing monotonicity of the LCE, see [Theorem 4.7\(a\)](#): $\|U_k\|_{\mathcal{G}} \leq Y$ a.s. does not imply that $\|\mathbb{E}^\mathbb{A}[U_k|V]\|_{\mathcal{G}} \leq Y$ a.s.

For a counterexample to the DCT under assumption (β) (without the finite-rank assumption) consider a centered Gaussian random variable V on $\mathcal{H} = \ell^2$ with Karhunen–Loève expansion $V = \sum_n \sigma_n Z_n e_n$, where $\sigma_n > 0$ for all $n \in \mathbb{N}$, $\sum_n \sigma_n^2 < \infty$, $Z_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$ and $(e_n)_{n \in \mathbb{N}}$ is the canonical basis of $\mathcal{H} = \ell^2$. Choose $\mathcal{G} = \mathbb{R}$ and $U_k = \delta_k Z_k$ where $\delta_k \searrow 0$ such that $\mathbb{P}[A_k] = 1/k$ for $A_k = [\delta_k Z_k \geq \varepsilon]$ and $\varepsilon = 1$. Then $\mathbb{E}^\mathbb{A}[U_k|V] = U_k$ by definition of the LCE and, since the family of events $(A_k)_{k \in \mathbb{N}}$ is independent and $\sum_k \mathbb{P}[A_k] = \infty$, the second Borel–Cantelli lemma implies $\mathbb{P}[A_k \text{ i.o.}] = 1$. Hence, by [Brémaud \(2017, Theorem 4.1.6\)](#), $\mathbb{E}^\mathbb{A}[U_k|V] = U_k$ does not converge to zero a.s., while (β) is satisfied since $\delta_k \rightarrow 0$.

It is also worth mentioning that the LCE does *not* fulfil several important properties of conditional expectations. These are summarised by the following statements and the actual counterexamples are provided in the proof in [Section 8](#). Note in particular that there are scalar-valued counterexamples in each case, and so these deficiencies of the LCE are not merely a consequence of the Hilbert space context.

Theorem 4.7 (Basic properties **not** satisfied by the LCE and the LCC). *For each of the following desired properties of the LCE, there is an explicit counterexample using $\mathbb{R}$ -valued random variables U, U', V , etc. satisfying the assumptions of [Theorem 4.5](#). As usual, all statements have to be understood $\mathbb{P}$ -a.s.*

- (a) **monotonicity:** *(invalid)*
 $U \geq U'$ implies $\mathbb{E}^\mathbb{A}[U|V] \geq \mathbb{E}^\mathbb{A}[U'|V]$ in the case $\mathcal{G} = \mathbb{R}$;
- (b) **triangle inequality:** *(invalid)*
 $\|\mathbb{E}^\mathbb{A}[U|V]\|_{\mathcal{G}} \leq \mathbb{E}^\mathbb{A}[\|U\|_{\mathcal{G}}|V]$;
- (c) **Jensen's inequality:** *(invalid)*
 $f(\mathbb{E}^\mathbb{A}[U|V]) \leq \mathbb{E}^\mathbb{A}[f(U)|V]$ for any convex³ function $f: \mathcal{G} \rightarrow \mathbb{R}$;
- (d) **pulling out known factors:** *(invalid)*
 $\mathbb{E}^\mathbb{A}[f(V)U|V] = f(V)\mathbb{E}^\mathbb{A}[U|V]$ for measurable⁴ maps $f: \mathcal{H} \rightarrow \mathbb{R}$;
- (e) **(yet another) tower property:** *(invalid)*
 $\mathbb{E}^\mathbb{A}[\mathbb{E}^\mathbb{A}[U|V] | W] = \mathbb{E}^\mathbb{A}[U|W]$ if $\sigma(W) \subseteq \sigma(V)$;
- (f) **Fatou's lemma:** *(invalid)*
 Let $\mathcal{G} = \mathbb{R}$ and $\mathbb{E}^\mathbb{A}[\inf_{k \in \mathbb{N}} U_k|V] < \infty$ (alternatively, $\mathbb{E}[\inf_{k \in \mathbb{N}} U_k|V] < \infty$). Then
 $\mathbb{E}^\mathbb{A}[\liminf_{k \rightarrow \infty} U_k|V] \leq \liminf_{k \rightarrow \infty} \mathbb{E}^\mathbb{A}[U_k|V]$;
- (g) **L^1 -dominated convergence theorem:** *(invalid)*

³Note that Jensen's (in-)equality holds for affine functions $f \in \mathcal{A}(\mathcal{G}; \mathcal{F})$, cf. [Theorem 4.5\(b\)](#).

⁴Our counterexample shows that property (d) is invalid even if “measurable” is strengthened to “linear”.

2 The only external ingredient

The pointwise pathology needed below is classical.

Lemma 1 (Menshov). *There are a real orthonormal system $(\varphi_n)_{n \geq 1}$ in $L^2([0,1])$ and real numbers $(c_n)_{n \geq 1}$ such that*

$$\sum_{n=1}^{\infty} c_n^2 < \infty, \quad \sum_{n=1}^N c_n \varphi_n(x) \quad \text{fails to converge as } N \rightarrow \infty$$

for every $x \in [0,1]$, after changes on a null set if necessary.

Proof. Menshov proved that for every positive $W(n) = o((\log n)^2)$, there are an orthonormal system and coefficients whose orthogonal series diverges everywhere although $\sum_n c_n^2 W(n) < \infty$ [2]. Take $W \equiv 1$. $\square$

We will also need mean-zero functions. Let

$$X_1 = [0, 1] \times \{-1, 1\}, \quad m_1 = \lambda \otimes \frac{1}{2}(\delta_{-1} + \delta_1),$$

and set

$$\psi_n(x, \varepsilon) = \varepsilon \varphi_n(x). \quad (3)$$

Then (ψ_n) is orthonormal in $L^2(X_1, m_1)$, every ψ_n has mean zero, and the series $\sum c_n \psi_n$ diverges at every point where the Menshov series does.

Nevertheless, Riesz–Fischer gives an L^2 sum F of this series. If

$$r_k = F - \sum_{n < k} c_n \psi_n, \quad (4)$$

then r_k is the L^2 tail, but $r_k \not\rightarrow 0$ almost everywhere; otherwise the finite partial sums would converge pointwise to F .

3 A graph projection that destroys dominated convergence

Let $X_0 = [0, 1]$ with Lebesgue measure. Choose pairwise disjoint intervals $I_n \subset X_0$, split each into equal halves $I_n = I_n^+ \dot{\cup} I_n^-$, and define

$$\delta_n = |I_n|^{-1/2}(\mathbf{1}_{I_n^+} - \mathbf{1}_{I_n^-}). \quad (5)$$

The functions δ_n are orthonormal, mean zero, and have disjoint supports. Define the isometry

$$T : \overline{\text{span}}\{\delta_n\} \longrightarrow \overline{\text{span}}\{\psi_n\}, \quad T\delta_n = \psi_n. \quad (6)$$

Form the disjoint-union probability space

$$(\Omega, \mathbb{P}) = \frac{1}{2}(X_0, m_0) \dot{\cup} \frac{1}{2}(X_1, m_1).$$

Thus $L^2(\Omega)$ is the half-weighted direct sum $L^2(X_0) \oplus L^2(X_1)$. Put

$$s_n = (\delta_n, \psi_n), \quad \mathcal{S} = \overline{\text{span}}\{s_n : n \geq 1\}. \quad (7)$$

Because of the half weights, (s_n) is an orthonormal system. The subspace $\mathcal{S}$ is the graph of T with this normalization, and direct minimization gives

$$P_{\mathcal{S}}(f, 0) = \frac{1}{2}(f, Tf) \quad \text{for } f \in \overline{\text{span}}\{\delta_n\}. \quad (8)$$

Indeed, if $f = \sum a_n \delta_n$, then $\langle (f, 0), s_n \rangle_{L^2(\Omega)} = a_n/2$, which proves (8).

Now define the disjoint-support tails

$$f_k = \sum_{n \geq k} c_n \delta_n, \quad U_k = (f_k, 0). \quad (9)$$

These are genuine pointwise sums: at each $x \in X_0$, at most one term is nonzero. In particular, $f_k(x) \rightarrow 0$ for every x outside the interval endpoints. Moreover,

$$Y = \left(\sum_{n \geq 1} |c_n| |\delta_n|, 0 \right) \quad (10)$$

dominates every U_k , and disjointness gives

$$\|Y\|_{L^2(\Omega)}^2 = \frac{1}{2} \sum_{n \geq 1} c_n^2 < \infty. \quad (11)$$

Thus $U_k \rightarrow 0$ almost surely under one square-integrable dominator.

On the other hand, $Tf_k = r_k$ as L^2 equivalence classes: both are the tail of the same orthogonal series. Hence (8) and (4) give

$$P_S U_k = \frac{1}{2}(f_k, r_k), \quad (12)$$

which does not converge to zero almost surely on the second component of Ω .

4 Realizing the graph projection as an infinite-rank LCE

It remains to realize $\mathbb{R}\mathbf{1} \oplus \mathcal{S}$ as the affine coordinate space of one Hilbert-valued random variable. Let (e_n) be the standard basis of ℓ^2 , choose $\tau_n = 2^{-n}$, and define

$$V(\omega) = \sum_{n=1}^{\infty} \tau_n s_n(\omega) e_n. \quad (13)$$

Tonelli's theorem and orthonormality show that

$$\mathbb{E}\|V\|_{\ell^2}^2 = \sum_n \tau_n^2 < \infty,$$

so (13) defines an ℓ^2 -valued square-integrable random variable almost surely. Since both δ_n and ψ_n have mean zero, $\mathbb{E}V = 0$. Also

$$\langle C_V e_m, e_n \rangle = \mathbb{E}[\tau_m s_m \tau_n s_n] = \tau_n^2 \mathbf{1}_{m=n}. \quad (14)$$

Thus $C_V = \text{diag}(4^{-n})$ has infinite rank.

For $h = (h_n) \in \ell^2$, a bounded scalar linear functional of V is

$$\langle h, V \rangle = \sum_n h_n \tau_n s_n.$$

Every finite linear combination of the s_n occurs in this way because all τ_n are nonzero. Consequently,

$$\overline{\{b + \langle h, V \rangle : b \in \mathbb{R}, h \in \ell^2\}} = \mathbb{R}\mathbf{1} \oplus \mathcal{S}. \quad (15)$$

The sum is orthogonal because every s_n has mean zero. Each U_k also has mean zero, so (1), (12), and (15) imply

$$\mathbb{E}^{\mathcal{A}}[U_k \mid V] = P_S U_k = \frac{1}{2}(f_k, r_k) \not\rightarrow 0 \quad \text{a.s.}$$

This proves the announced counterexample.

5 Conclusion and scope

The source’s finite-rank hypothesis cannot be removed from its L^2 -dominated convergence theorem, even when the predicted variables are real-valued and the conditioning variable is centered and separable. The mechanism is not the unbounded covariance pseudoinverse: it is the fact that, at infinite rank, the LCE subspace can encode an arbitrary separable orthogonal projection, including one that transports harmless disjoint tails to the pointwise-divergent tails of an orthogonal series.

The finite-dimensional graph projection and covariance algebra were checked by the accompanying script. The only nonconstructive input is the classical, sharp Menshov divergence theorem; all probabilistic and LCE steps are explicit.

References

- [1] I. Klebanov, B. Sprungk, and T. J. Sullivan, *The linear conditional expectation in Hilbert space*, arXiv:2008.12070, Remark 4.6.
- [2] D. Menchoff, *Sur les séries de fonctions orthogonales*, Fundamenta Mathematicae **4** (1923), 82–105, doi:10.4064/fm-4-1-82-105.